

MSIT — STUDENT LAPTOP SPECIFICATION

Purpose. MSIT coursework will involve programming, data science, machine learning, AI-assisted development, notebooks, containers, databases, collaborative tools and local prototyping. Heavy GPU training is expected to use institute or cloud compute; therefore, a dedicated laptop GPU is useful but not mandatory.

Component	Minimum	Recommended	High-End	Why this matters for MSIT
Processor	Intel Core i5 / AMD Ryzen 5 / Apple M2-class or newer	Intel Core Ultra 5/7, AMD Ryzen 7, or Apple M3/M4-class	Intel Core Ultra 7/9, AMD Ryzen 9, or Apple M4 Pro-class / equivalent	A modern multi-core CPU keeps IDEs, notebooks, compilers, local services and AI-assisted development responsive when several tools are running together.
Memory (RAM)	16 GB	32 GB	48–64 GB	RAM is the most important practical constraint for running browsers, IDEs, Jupyter, Docker/containers, databases and local AI tools simultaneously. 32 GB is the preferred new-purchase baseline.
Storage	512 GB SSD	1 TB NVMe SSD	2 TB NVMe SSD	Datasets, virtual environments, container images, model checkpoints and project repositories consume storage quickly. SSD storage is required for acceptable development performance.
Graphics	Integrated graphics or Apple Silicon integrated GPU	Integrated graphics / Apple Silicon GPU; NVIDIA RTX 4060-class optional for ML-focused students	NVIDIA RTX 4060/4070-class or better (preferably 8 GB+ VRAM), or higher-end Apple Silicon GPU	Integrated graphics and Apple Silicon GPUs are sufficient for most coursework and local ML inference. NVIDIA CUDA-capable GPUs can accelerate local deep-learning experiments but are not required for the programme. Larger training workloads should use institute or cloud GPUs.
Display	14–15.6 inch; Full HD (1920×1080) or equivalent	14–16 inch; 1920×1200 or higher; IPS/OLED	14–16 inch high-resolution IPS/OLED panel	Students spend long periods reading code, notebooks, terminals and documentation. A sharp, comfortable display improves productivity and reduces eye strain.
Battery	At least 6 hours of practical use	8–10+ hours of practical use	10+ hours of practical use	Students will frequently move between classrooms, labs, meetings and project spaces. Good battery life reduces dependence on charging points during the day.

Component	Minimum	Recommended	High-End	Why this matters for MSIT
Connectivity	Wi-Fi 6; Bluetooth 5.x	Wi-Fi 6E or Wi-Fi 7; Bluetooth 5.x	Wi-Fi 7; latest Bluetooth	Reliable wireless connectivity is important for cloud notebooks, Git repositories, video calls, remote development and access to institute services.
Ports	USB-C plus at least one practical external-display / peripheral option	USB-C / USB4 / Thunderbolt; HDMI; USB-A preferred	Multiple USB-C/Thunderbolt/USB4 ports, HDMI and convenient peripheral support	Students regularly connect external displays, storage, presentation equipment and development peripherals. Adapter-only configurations are usable but less convenient.
Webcam & audio	720p webcam and functional microphone	1080p webcam with good microphone quality	1080p+ webcam with high-quality microphone array	Useful for mentoring, remote collaboration, interviews, project reviews and industry co-op meetings.
Operating system	Windows 11, macOS, or a Linux-capable system	Windows 11 or macOS with reliable access to Linux tools/containers	Same; chosen according to the student's development workflow	MSIT work will rely heavily on cross-platform development tools. Students should be able to use Python, Git, containers and Linux-oriented workflows. CUDA-specific work may require remote NVIDIA compute on macOS.
Weight / portability	Preferably below 2.2 kg	Approximately 1.4–1.8 kg	Approximately 1.3–1.8 kg	The laptop is expected to be carried daily across classes, project work and co-op activities; portability has a meaningful impact on everyday use.
Warranty	1 year manufacturer warranty	3-year warranty recommended	3-year warranty plus accidental-damage coverage where available	The laptop is a primary academic tool. Multi-year coverage reduces the risk of extended downtime during the programme.

Recommended Purchase Profile

For most students buying a new laptop for MSIT, the recommended target is: **modern Core Ultra / Ryzen 7 / Apple M3–M4 class processor, 32 GB RAM, 1 TB SSD, 14–16 inch Full HD+ display, Wi-Fi 6 or newer, and a 3-year warranty.**

Minimum	Recommended	High-End
Suitable for an existing recent laptop. A student buying a new machine should preferably choose the Recommended tier rather than target the minimum.	The best balance of cost, portability and useful life for typical MSIT coursework. 32 GB RAM and a 1 TB SSD are strongly preferred for a new purchase.	Appropriate for students who expect substantial local ML/AI experimentation, large development environments, or who want a longer usable life. It is not required for normal coursework.

Platform guidance

- **Windows / Linux laptops:** A dedicated NVIDIA GPU is optional. It is useful for local CUDA experiments but should not be treated as a programme requirement.
- **Apple Silicon Macs:** Suitable for programming, data science and local AI inference. For a new purchase, 24–32 GB unified memory and 1 TB storage are preferable. CUDA-specific exercises can be run on remote/institute GPU systems.
- **Cloud / institute compute:** Students should expect to use remote compute for large deep-learning models, GPU-intensive training and workloads that exceed practical laptop limits.
- **Upgradeability:** Where possible, choose systems with adequate RAM and SSD capacity at purchase time. Many modern laptops do not allow memory upgrades later.

Important: These are capability guidelines, not brand recommendations. Equivalent or better configurations from any reputable manufacturer are acceptable. Specific software requirements may evolve during the programme as tools and platforms change.